Learning University Mathematics through the Algebra of Bookkeeping

kiwamizamurai

Table of contents

Preface 1
- Why Part 0 comes first . . . 1
- A rule this book follows . . . 1

I. Part 0: Research Methods 3

1. Asking a Researchable Question 7
- 1.1. From an interest to questions . . . 7
- 1.2. Deciding the scope . . . 7
- 1.3. Writing the question down . . . 8
- 1.4. Exercises . . . 8

2. Collecting Keywords 9
- 2.1. Each field has its own name for the idea . . . 9
- 2.2. Where the words came from . . . 10
- 2.3. Keywords in a second language . . . 10
- 2.4. Exercises . . . 10

3. Finding the First Paper 11
- 3.1. The first query . . . 11
- 3.2. One paper, several versions . . . 11
- 3.3. Checking the bibliographic data . . . 12
- 3.4. Exercises . . . 12

4. Following Citations 13
- 4.1. Backward and forward . . . 13
- 4.2. Similarity graphs: Connected Papers . . . 13
- 4.3. Database APIs: Semantic Scholar and OpenAlex . . . 14
- 4.4. Exercises . . . 14

5. Searching Literature in Japanese 15
- 5.1. The databases . . . 15
- 5.2. A worked example: one term, two tools . . . 15
- 5.3. Reading a title in another language . . . 16
- 5.4. Exercises . . . 16

6. Finding Older Literature **17**
6.1. Digitized books: the Internet Archive . . . 17
6.2. A nineteenth-century article . . . 17
6.3. History papers as guides . . . 17
6.4. Exercises . . . 18

7. Reading Papers **19**
7.1. Read in passes . . . 19
7.2. Extracting the main claim . . . 19
7.3. Reading a paper in a field you do not know . . . 20
7.4. Exercises . . . 20

8. Organizing What You Find **21**
8.1. The research log . . . 21
8.2. The bibliography file . . . 21
8.3. A map of research lines . . . 21
8.4. Exercises . . . 22

9. Limits and Pitfalls **23**
9.1. Usage limits . . . 23
9.2. Gaps in documentation . . . 23
9.3. A registry match that was a different record . . . 23
9.4. Copies from sites that are not the publisher . . . 24
9.5. Fabricated references from chatbots . . . 24
9.6. Our own near-misses . . . 24
9.7. Exercises . . . 25

10. Writing Up the Survey **27**
10.1. A timeline . . . 27
10.2. Where this book stands . . . 28
10.3. Exercises . . . 28

II. Part I: The Language of Mathematics and Proof **29**

11. Propositions and Logic **33**
11.1. Planned sections . . . 33
11.2. Exercises . . . 33
11.3. Formalization . . . 33

12. Writing Proofs **35**
12.1. Planned sections . . . 35
12.2. Exercises . . . 35
12.3. Formalization . . . 35

13. Sets **37**
13.1. Planned sections . . . 37
13.2. Exercises . . . 37
13.3. Formalization . . . 37

14. Maps **39**
14.1. Planned sections . . . 39
14.2. Exercises . . . 39
14.3. Formalization . . . 39

15. Relations and Equivalence Relations **41**
15.1. Planned sections . . . 41
15.2. Exercises . . . 41
15.3. Formalization . . . 41

16. Orders **43**
16.1. Planned sections . . . 43
16.2. Exercises . . . 43
16.3. Formalization . . . 43

17. Induction and Recursion **45**
17.1. Planned sections . . . 45
17.2. Exercises . . . 45
17.3. Formalization . . . 45

18. Number Systems **47**
18.1. Planned sections . . . 47
18.2. Exercises . . . 47
18.3. Formalization . . . 47

III. Part II: Bookkeeping Basics **49**

19. What Bookkeeping Is **53**
19.1. Planned sections . . . 53
19.2. Exercises . . . 53

20. The Accounting Equation **55**
20.1. Planned sections . . . 55
20.2. Exercises . . . 55

21. Transactions and Journal Entries **57**
21.1. Planned sections . . . 57
21.2. Exercises . . . 57

22. Accounts and Posting **59**
22.1. Planned sections . . . 59
22.2. Exercises . . . 59

23. The Trial Balance **61**
23.1. Planned sections . . . 61
23.2. Exercises . . . 61

24. Closing the Books **63**
24.1. Planned sections . . . 63
24.2. Exercises . . . 63

25. Financial Statements **65**
25.1. Planned sections . . . 65
25.2. Exercises . . . 65

26. The Running Example **67**
26.1. Planned sections . . . 67
26.2. Exercises . . . 67

IV. Part III: Monoids and Groups **69**

27. Binary Operations **73**
27.1. Planned sections . . . 73
27.2. Exercises . . . 73
27.3. Formalization . . . 73

28. Monoids **75**
28.1. Planned sections . . . 75
28.2. Exercises . . . 75
28.3. Formalization . . . 75

29. Groups **77**
29.1. Planned sections . . . 77
29.2. Exercises . . . 77
29.3. Formalization . . . 77

30. Subgroups and Cosets **79**
30.1. Planned sections . . . 79
30.2. Exercises . . . 79
30.3. Formalization . . . 79

31. Homomorphisms **81**
31.1. Planned sections . . . 81

31.2. Exercises . . . 81
31.3. Formalization . . . 81

32. Quotient Groups **83**
32.1. Planned sections . . . 83
32.2. Exercises . . . 83
32.3. Formalization . . . 83

33. Direct Products and Direct Sums **85**
33.1. Planned sections . . . 85
33.2. Exercises . . . 85
33.3. Formalization . . . 85

34. Free Abelian Groups **87**
34.1. Planned sections . . . 87
34.2. Exercises . . . 87
34.3. Formalization . . . 87

V. Part IV: The Group of Differences **89**

35. T-Accounts as Pairs **93**
35.1. Planned sections . . . 93
35.2. Exercises . . . 93
35.3. Formalization . . . 93

36. The Pacioli Group **95**
36.1. Planned sections . . . 95
36.2. Exercises . . . 95
36.3. Formalization . . . 95
36.4. Further reading . . . 95

37. Journal Entries and the Trial Balance **97**
37.1. Planned sections . . . 97
37.2. Exercises . . . 97
37.3. Formalization . . . 97
37.4. Further reading . . . 97

38. The Accounting Equation, Algebraically **99**
38.1. Planned sections . . . 99
38.2. Exercises . . . 99
38.3. Formalization . . . 99

39. The Grothendieck Group **101**
39.1. Planned sections . . . 101

39.2. Exercises . . . 101
39.3. Formalization . . . 101

40. The Monoid of Records and the Group of Balances **103**
40.1. Planned sections . . . 103
40.2. Exercises . . . 103
40.3. Formalization . . . 103
40.4. Further reading . . . 103

41. Multi-Dimensional Bookkeeping **105**
41.1. Planned sections . . . 105
41.2. Exercises . . . 105
41.3. Formalization . . . 105
41.4. Further reading . . . 105

42. An Algebra of Errors **107**
42.1. Planned sections . . . 107
42.2. Exercises . . . 107
42.3. Formalization . . . 107

VI. Part V: Linear Algebra **109**

43. Vector Spaces **113**
43.1. Planned sections . . . 113
43.2. Exercises . . . 113
43.3. Formalization . . . 113

44. Matrices and Linear Maps **115**
44.1. Planned sections . . . 115
44.2. Exercises . . . 115
44.3. Formalization . . . 115

45. Matrix Bookkeeping **117**
45.1. Planned sections . . . 117
45.2. Exercises . . . 117
45.3. Formalization . . . 117
45.4. Further reading . . . 117

46. Kernel, Image, and Rank **119**
46.1. Planned sections . . . 119
46.2. Exercises . . . 119
46.3. Formalization . . . 119

47. Systems of Linear Equations **121**
47.1. Planned sections . . . 121
47.2. Exercises . . . 121
47.3. Formalization . . . 121

48. Least Squares and Generalized Inverses **123**
48.1. Planned sections . . . 123
48.2. Exercises . . . 123
48.3. Formalization . . . 123
48.4. Further reading . . . 123

49. Inner Products and Duality **125**
49.1. Planned sections . . . 125
49.2. Exercises . . . 125
49.3. Formalization . . . 125

50. Aggregation and Projection **127**
50.1. Planned sections . . . 127
50.2. Exercises . . . 127
50.3. Formalization . . . 127
50.4. Further reading . . . 127

51. Eigenvalues and Diagonalization **129**
51.1. Planned sections . . . 129
51.2. Exercises . . . 129
51.3. Formalization . . . 129

52. Input-Output Analysis **131**
52.1. Planned sections . . . 131
52.2. Exercises . . . 131
52.3. Formalization . . . 131

VII. Part VI: Rings, Modules, and Order **133**

53. Rings and Fields **137**
53.1. Planned sections . . . 137
53.2. Exercises . . . 137
53.3. Formalization . . . 137

54. Modules **139**
54.1. Planned sections . . . 139
54.2. Exercises . . . 139
54.3. Formalization . . . 139
54.4. Further reading . . . 139

55. Tensor Products **141**
55.1. Planned sections . . . 141
55.2. Exercises . . . 141
55.3. Formalization . . . 141

56. Ordered Groups **143**
56.1. Planned sections . . . 143
56.2. Exercises . . . 143
56.3. Formalization . . . 143

57. Lattices and the Chart of Accounts **145**
57.1. Planned sections . . . 145
57.2. Exercises . . . 145
57.3. Formalization . . . 145

VIIIPart VII: Graphs and Networks **147**

58. Graphs **151**
58.1. Planned sections . . . 151
58.2. Exercises . . . 151
58.3. Formalization . . . 151

59. The Incidence Matrix **153**
59.1. Planned sections . . . 153
59.2. Exercises . . . 153
59.3. Formalization . . . 153

60. Flows and Conservation **155**
60.1. Planned sections . . . 155
60.2. Exercises . . . 155
60.3. Formalization . . . 155

61. Trees and Spanning Trees **157**
61.1. Planned sections . . . 157
61.2. Exercises . . . 157
61.3. Formalization . . . 157

62. Cycle Space and Cut Space **159**
62.1. Planned sections . . . 159
62.2. Exercises . . . 159
62.3. Formalization . . . 159

63. An Introduction to Graph Homology **161**
63.1. Planned sections . . . 161

63.2. Exercises . . . 161
63.3. Formalization . . . 161

64. Bookkeeping Graphs **163**
64.1. Planned sections . . . 163
64.2. Exercises . . . 163
64.3. Formalization . . . 163
64.4. Further reading . . . 163

IX. Part VIII: Category Theory **165**

65. Categories **169**
65.1. Planned sections . . . 169
65.2. Exercises . . . 169
65.3. Formalization . . . 169

66. Functors and Natural Transformations **171**
66.1. Planned sections . . . 171
66.2. Exercises . . . 171
66.3. Formalization . . . 171

67. Universal Properties **173**
67.1. Planned sections . . . 173
67.2. Exercises . . . 173
67.3. Formalization . . . 173

68. Adjunctions **175**
68.1. Planned sections . . . 175
68.2. Exercises . . . 175
68.3. Formalization . . . 175

69. Monoidal Categories and String Diagrams **177**
69.1. Planned sections . . . 177
69.2. Exercises . . . 177
69.3. Formalization . . . 177

70. Spans and Cospans **179**
70.1. Planned sections . . . 179
70.2. Exercises . . . 179
70.3. Formalization . . . 179
70.4. Further reading . . . 179

X. Part IX: Time and Change 181

71. Sequences and Difference Equations 185
71.1. Planned sections . . . 185
71.2. Exercises . . . 185
71.3. Formalization . . . 185

72. Series and Present Value 187
72.1. Planned sections . . . 187
72.2. Exercises . . . 187
72.3. Formalization . . . 187

73. Derivatives and Integrals 189
73.1. Planned sections . . . 189
73.2. Exercises . . . 189
73.3. Formalization . . . 189

74. Differential Equations 191
74.1. Planned sections . . . 191
74.2. Exercises . . . 191
74.3. Formalization . . . 191

75. Triple-Entry Bookkeeping 193
75.1. Planned sections . . . 193
75.2. Exercises . . . 193
75.3. Formalization . . . 193
75.4. Further reading . . . 193

76. Probability and Estimates 195
76.1. Planned sections . . . 195
76.2. Exercises . . . 195
76.3. Formalization . . . 195

XI. Part X: Computation and Information 197

77. Automata 201
77.1. Planned sections . . . 201
77.2. Exercises . . . 201
77.3. Formalization . . . 201
77.4. Further reading . . . 201

78. The Ledger as a Data Structure 203
78.1. Planned sections . . . 203
78.2. Exercises . . . 203

78.3. Formalization . . . 203

79. Relational Databases **205**
79.1. Planned sections . . . 205
79.2. Exercises . . . 205
79.3. Formalization . . . 205

80. Plain-Text Accounting **207**
80.1. Planned sections . . . 207
80.2. Exercises . . . 207
80.3. Formalization . . . 207

81. Error Detection **209**
81.1. Planned sections . . . 209
81.2. Exercises . . . 209
81.3. Formalization . . . 209

82. Types That Rule Out Unbalanced Entries **211**
82.1. Planned sections . . . 211
82.2. Exercises . . . 211
82.3. Formalization . . . 211

XII. Part XI: Formalizing Bookkeeping in Lean **213**

83. Lean 4 Basics **217**
83.1. Planned sections . . . 217
83.2. Exercises . . . 217
83.3. Formalization . . . 217

84. Using Mathlib **219**
84.1. Planned sections . . . 219
84.2. Exercises . . . 219
84.3. Formalization . . . 219

85. Defining Accounts and Journal Entries **221**
85.1. Planned sections . . . 221
85.2. Exercises . . . 221
85.3. Formalization . . . 221

86. Building the Pacioli Group **223**
86.1. Planned sections . . . 223
86.2. Exercises . . . 223
86.3. Formalization . . . 223

87. Proving That the Trial Balance Balances **225**
87.1. Planned sections . . . 225
87.2. Exercises . . . 225
87.3. Formalization . . . 225

88. Comparing with the Grothendieck Group in Mathlib **227**
88.1. Planned sections . . . 227
88.2. Exercises . . . 227
88.3. Formalization . . . 227

89. Matrix Bookkeeping in Lean **229**
89.1. Planned sections . . . 229
89.2. Exercises . . . 229
89.3. Formalization . . . 229

90. Graphs and Incidence Matrices in Lean **231**
90.1. Planned sections . . . 231
90.2. Exercises . . . 231
90.3. Formalization . . . 231

91. Checking the Running Example in Lean **233**
91.1. Planned sections . . . 233
91.2. Exercises . . . 233
91.3. Formalization . . . 233

92. Running a Formalization Project **235**
92.1. Planned sections . . . 235
92.2. Exercises . . . 235
92.3. Formalization . . . 235

XIIIPart XII: Applications in Accounting **237**

93. Consolidation **241**
93.1. Planned sections . . . 241
93.2. Exercises . . . 241
93.3. Formalization . . . 241

94. Foreign Currency Translation **243**
94.1. Planned sections . . . 243
94.2. Exercises . . . 243
94.3. Formalization . . . 243

95. Cost Accounting **245**
95.1. Planned sections . . . 245

95.2. Exercises . . . 245
95.3. Formalization . . . 245

96. Management Accounting and Optimization **247**
96.1. Planned sections . . . 247
96.2. Exercises . . . 247
96.3. Formalization . . . 247

97. The Cash Flow Statement, Algebraically **249**
97.1. Planned sections . . . 249
97.2. Exercises . . . 249
97.3. Formalization . . . 249
97.4. Further reading . . . 249

98. Auditing and Anomaly Detection **251**
98.1. Planned sections . . . 251
98.2. Exercises . . . 251
98.3. Formalization . . . 251

XIV Part XIII: Applications in Economics **253**

99. National Accounts **257**
99.1. Planned sections . . . 257
99.2. Exercises . . . 257
99.3. Formalization . . . 257

100 Input-Output Tables **259**
100.1 Planned sections . . . 259
100.2 Exercises . . . 259
100.3 Formalization . . . 259

101 Social Accounting Matrices **261**
101.1 Planned sections . . . 261
101.2 Exercises . . . 261
101.3 Formalization . . . 261

XV. Part XIV: History and Research Directions **263**

102 The Origins of Double Entry **267**
102.1 Planned sections . . . 267
102.2 Exercises . . . 267
102.3 Further reading . . . 267

103 Negative Numbers and Bookkeeping **269**
103.1 Planned sections . . . 269
103.2 Exercises . . . 269
103.3 Further reading . . . 269

104 The Algebra of Accounts, from Sprague Onward **271**
104.1 Planned sections . . . 271
104.2 Exercises . . . 271
104.3 Further reading . . . 271

105 Research in Japanese **273**
105.1 Planned sections . . . 273
105.2 Exercises . . . 273
105.3 Further reading . . . 273

106 Open Problems and Thesis Topics **275**
106.1 Planned sections . . . 275
106.2 Exercises . . . 275

Appendices **277**

A. Notation **277**
A.1. Planned sections . . . 277

B. Glossary **279**
B.1. Planned sections . . . 279

C. The Running Example in Full **281**
C.1. Planned sections . . . 281

D. Lean Tactic Reference **283**
D.1. Planned sections . . . 283

E. Solutions to Exercises **285**
E.1. Planned sections . . . 285

References **287**

Preface

This book teaches university mathematics through one running example: double-entry bookkeeping. Each part takes the same small set of business transactions and describes them again in a new mathematical language. Sets and maps come first, then monoids and groups, vector spaces, graphs, and categories. The main theorems are formalized in Lean 4, so a reader can check them by machine.

The book does not assume any knowledge of bookkeeping. It assumes high-school mathematics.

Why Part 0 comes first

Before any mathematics, Part 0 shows how the material in this book was found. It treats "the algebra of bookkeeping" as a research topic and walks through the search that produced this book's bibliography: the queries, the databases, the tools that worked, and the tools that failed. Every search reported in Part 0 was carried out on 15 October 2026, and the results describe what the tools returned on that day. Search results change over time, so a reader who repeats a search should expect different numbers.

A rule this book follows

Every factual claim in this book is checked against a source before it is written down, and a claim that cannot be checked is left out. Bibliographic data was checked against Crossref, arXiv, CiNii Research, Internet Archive catalog records, publishers' pages, or the papers themselves.

Part I.

Part 0: Research Methods

The rest of this book teaches mathematics through bookkeeping. Part 0 comes first because it answers a question a careful reader will ask: where does this material come from, and how do we know it is right?

Part 0 follows one literature search from start to finish. The topic is the algebra of bookkeeping itself, so the papers it finds are the sources of the later parts. Chapter 1 to Chapter 7 show how to find and read those sources. Chapter 8 and Chapter 10 show how to record them and summarize them. Chapter 9 collects the places where tools, records, and our own drafts went wrong.

Each chapter ends with exercises that apply the same steps to a topic of your choice. A reader who wants to start the mathematics at once can go to Part I and come back to Part 0 when a later chapter cites a source.

1. Asking a Researchable Question

Every search starts from a question. A vague question produces a vague search, so the first task is to turn a loose interest into questions that a database can answer.

1.1. From an interest to questions

This book began with a loose interest: *what is double-entry bookkeeping, mathematically?* A database cannot answer that sentence directly. Looking back at our search, it answered the following smaller questions, each of which names something a database can find. Writing such questions down at the start, as we recommend here, makes the search easier to plan.

1. Has anyone described double-entry bookkeeping as an algebraic structure, such as a group or a monoid?
2. Has anyone written bookkeeping with matrices or vectors?
3. Has anyone described bookkeeping as a graph or a network?
4. Has anyone used category theory for bookkeeping?
5. Has anyone stated axioms for accounting measurement?
6. When did these ideas appear, and who wrote them first?

Each question points to a branch of mathematics, and each branch has its own vocabulary. Chapter 2 turns these questions into search terms.

1.2. Deciding the scope

A question also needs boundaries. Our search had four.

- **Fields.** The work could come from accounting, mathematics, computer science, or economics. We kept all four in scope, because a paper on "ledgers" in computer science may answer the same question as a paper on "accounts" in accounting.
- **Languages.** We searched in English and in Japanese. Part of the literature on this topic is written in Japanese (Chapter 5).
- **Time.** We set no lower limit on the publication year. According to Ellerman (2014), the double-entry method was in business use for over five centuries, and the oldest work in this book's bibliography dates from 1880 (Chapter 6).

- **Kind of source.** We accepted journal articles, preprints, books, and university repository papers. We excluded copies of works posted by sites that are not the publisher or the author (Chapter 9).

1.3. Writing the question down

Write the question and the scope in a research log before your first search. The log is a plain text file with dates. Each search later adds an entry: the date, the tool, the exact query, and what came back. The worked examples in this part describe searches we ran on 15 October 2026.

1.4. Exercises

1. Choose a topic you know a little about. Write one loose question about it, then split it into at least four questions that each name something a database could find.
2. For your topic, decide the four boundaries in this chapter: fields, languages, time, and kind of source. Give a reason for each decision.
3. Start a research log for your topic. Record today's date and the questions from Exercise 1.

2. Collecting Keywords

A search engine matches words. If the authors of a paper used a different word from yours, the search misses the paper. This chapter shows how we collected the words that authors in this field use.

2.1. Each field has its own name for the idea

The same idea appears under different names in different fields. These are the names we found in titles and abstracts during our search:

Name used by authors	Where we found it
double-entry bookkeeping	title of Ellerman (2014)
group of differences, Pacioli group	abstract of Ellerman (2014)
double entry multidimensional accounting	title of Ellerman (1986)
algebraic model of accounting systems	title of Nehmer and Robinson (1997)
algebraic automaton	title of Rambaud and Pérez (2005)
matrix-based accounting system	title of Leech (1986)
inferring transactions from financial statements	title of Arya et al. (2000)
bookkeeping graphs	title of Liang (2023)
partita doppia, spans of reflexive graphs	title and abstract of Katis et al. (2008)
triple-entry bookkeeping	title of Ijiri (1986)
簿記代数 (Japanese: “bookkeeping algebra”)	title of Ozawa (2023a)

Two lessons follow from this table.

First, the name of a mathematical structure can be a sharper search term than the name of the subject. The abstract of Ellerman (2014) introduces the name “Pacioli group” for the group of differences used in bookkeeping, so the term points to this line of work.

Second, the historical Italian name *partita doppia* is the title of a category theory paper (Katis et al. 2008). Its title does not contain the English words “double-entry bookkeeping,” so a search restricted to titles in English would miss it. In our search it appeared in response to a Google Scholar query that combined “double-entry,” “bookkeeping,” and “category theory.”

2.2. Where the words came from

We collected these terms in a loop:

1. Search with the words we already had.
2. Read the titles and abstracts of the results.
3. Add any new names for the idea to the list.
4. Search again with the new words.

The loop stops when new searches return only papers already in the log. In our search, the first query used the words “algebra,” “bookkeeping,” and “double-entry.” The term “Pacioli group” came from the abstract of the first relevant paper, and “bookkeeping graphs” came from a later query that added “graph theory.”

2.3. Keywords in a second language

For Japanese sources we used terms taken from Japanese titles. The term 簿記代数 (“bookkeeping algebra”) appears in a Japanese paper title (Ozawa 2023a), and searching for that exact term in a Japanese database found that paper directly (Chapter 5).

2.4. Exercises

1. For your topic from Chapter 1, list at least five names for the central idea. For each, record where you found it.
2. Run the loop in this chapter twice. Record in your log which new terms the second round added.
3. Find one term for your topic in a language other than English. Search for it, and compare the results with the English search.

3. Finding the First Paper

A literature search needs a starting point: one paper that is on topic. From that paper you can follow references and citations (Chapter 4). This chapter shows how we found ours and how we checked its bibliographic data.

3.1. The first query

Our first search used a general web search engine with the query

```
"algebra" bookkeeping double-entry book filetype:pdf
```

The `filetype:pdf` part asks for PDF files, and all but one of the results we received were PDFs. Among the results was a PDF of Ellerman (1985) on the author's own website, and a PDF of Ellerman (2014) on arXiv. Both are by the same author. The abstract of Ellerman (2014) describes bookkeeping in terms of a group, which matched our first question in Chapter 1. We chose Ellerman (2014) as the starting paper because it is the more recent of the two. A recent paper is a good start because its reference list can lead back to earlier work.

3.2. One paper, several versions

A paper often exists in more than one version. Ellerman (2014) has at least three records:

Version	Identifier	Date
arXiv preprint	arXiv:1407.1898	2014-07-07
SSRN posting	DOI 10.2139/ssrn.2463442	2014
Journal article in *Accounting Education*, vol. 23, no. 5, pp. 483–501	DOI 10.1080/09639284.2014.949803	2014

The arXiv date comes from the arXiv API. The other two rows come from the records registered with Crossref, which we queried through its public API. When you cite a paper, cite the published version, and record the preprint as a freely readable copy.

3.3. Checking the bibliographic data

A search result is a pointer to a paper, and the details it shows can be incomplete. Before a reference enters the bibliography of this book, we check it against a record kept by the publisher or by a registry.

1. **arXiv.** The arXiv API returns the title, authors, abstract, subject category, and first submission date of a preprint. We queried it with `au:Ellerman AND abs:bookkeeping` and received exactly one record, arXiv:1407.1898.
2. **Crossref.** We sent each journal article and conference paper in this book's bibliography, as free text, to Crossref's public API (`api.crossref.org/works` with the `query.bibliographic` parameter). It returned the closest registered records with their DOI, journal, volume, issue, pages, and date.

The Crossref result is only a candidate. Compare it with the reference you are checking: same authors, same title, same year. Chapter 9 shows a case where the best match was a different record.

3.4. Exercises

1. Find a starting paper for your topic. Explain why you chose it over the other results.
2. Find every version of your starting paper (preprint, journal article, repository copy). Record each version's identifier and date in your log.
3. Send the citation of your starting paper to the Crossref API and compare the returned record with what you have. Note any differences.

4. Following Citations

Once you have one good paper, the papers connected to it lead to the rest of the literature. This chapter covers three ways to follow those connections: by hand through references and citations, with a similarity graph, and with a scholarly database API.

4.1. Backward and forward

A paper connects to the literature in two directions.

- **Backward**: the works it cites, listed in its references. These are older.
- **Forward**: the later works that cite it. Google Scholar shows these under "Cited by."

Google Scholar's search help gives a rule of thumb for choosing a direction. If your results are too specific, look at the works they cite, which tend to be more general. If your results are too basic, look at the papers that cite them, which tend to be newer and more specific.

We repeated both steps from each new relevant paper. Wohlin (2014) gives guidelines for conducting literature reviews with a snowballing approach, evaluates them by replicating a published systematic review, and concludes that snowballing can be a good alternative to database searches as a first search strategy.

4.2. Similarity graphs: Connected Papers

Connected Papers draws a graph around one "origin" paper. According to its own description, it analyzes on the order of 50,000 papers and keeps the few dozen with the strongest connections to the origin. The connection measure is based on *co-citation* and *bibliographic coupling*: two papers whose citations and references overlap heavily are presumed more likely to treat related subject matter. Papers that never cite each other can therefore sit close together. The service states that its graph is not a citation tree. Its data comes from the Semantic Scholar paper corpus.

We built a graph with Ellerman (2014) as the origin. Among the papers it placed in the graph were

- earlier work by the same author: Ellerman (1985) and Ellerman (1986);

- Rambaud et al. (2010), a book on algebraic models for accounting systems;
- matrix accounting: Leech (1986);
- linear algebra applied to financial statements: Arya et al. (2000) and Arya et al. (2004);
- history: Peters and Emery (1978) and Stone (2021).

The same graph also contained papers on unrelated subjects, such as a strategic analysis of small and medium-sized accounting firms in Jing Zhou and a study of the social consequences of the indenture system in Fiji. A similarity graph measures overlap in citations, and overlap does not guarantee a shared topic. Read the title and abstract of every node before adding it to your log.

A second graph, with Rambaud et al. (2010) as the origin, added earlier algebraic work: Nehmer and Robinson (1997) and Rambaud and Pérez (2005).

4.3. Database APIs: Semantic Scholar and OpenAlex

For larger searches, a program can follow citations for you through a database API.

- **Semantic Scholar** offers the Academic Graph API. Most endpoints work without an API key, but unauthenticated requests share a rate limit of 1,000 requests per second across all such users and may be throttled further when traffic is heavy. A personal API key starts with a limit of one request per second.
- **OpenAlex** describes itself as a free and open catalog of the world's scholarly research system. Its data is released under the CC0 license and is available through an API and as a bulk download.

When we sent unauthenticated search requests to the Semantic Scholar API on 15 October 2026, it answered "Too Many Requests." Chapter 9 returns to this.

4.4. Exercises

1. Take your starting paper from Chapter 3. List five works it cites and five works that cite it.
2. Build a Connected Papers graph around your starting paper. For each node, decide from its title and abstract whether it belongs in your log. How many nodes did you reject?
3. Write a short program that queries the OpenAlex API for the works that cite your starting paper. Compare its list with Google Scholar's "Cited by" list.

5. Searching Literature in Japanese

Part of the work on the algebra of bookkeeping is written in Japanese. The papers in this chapter appeared in a university bulletin and are available from a university repository. This chapter introduces the Japanese databases we used and shows how their results differed.

5.1. The databases

- **CiNii Research**, run by the National Institute of Informatics (NII), searches several kinds of records from one box: data, articles, books, dissertations, research projects, and researchers.
- **IRDB** (Institutional Repositories DataBase) is also run by NII. As its name says, it is a database of institutional repositories.
- **J-STAGE** is a platform for scholarly publications in Japan, developed and managed by the Japan Science and Technology Agency (JST). Its overview page lists more than 4,000 journals, proceedings, and other publications from more than 2,400 publishers in Japan.
- **NDL Search**, run by the National Diet Library, searches the library's own printed and digital holdings. It also searches bibliographic and location information for books held by prefectural and municipal libraries, and materials from university and specialty libraries and research institutions across Japan.

5.2. A worked example: one term, two tools

The Japanese term 簿記代数 ("bookkeeping algebra") came from our keyword loop (Chapter 2): a Google Scholar query in Japanese (簿記 代数 構造 OR 複式簿記 数学 群) returned a paper whose title contains it. On 15 October 2026 we then searched two tools.

CiNii Research. A search for 簿記代数 across all record types returned one result: Ozawa (2023a), published in 2023 in the bulletin 商学研究科紀要, no. 97, pp. 79–96.

Google Scholar. A search for the author of that paper together with 複式簿記 ("double-entry bookkeeping") returned three papers by this author with PDFs in the Waseda University repository: Ozawa (2023a), Ozawa (2023b), and Ozawa (2026). It also listed one co-authored paper without a PDF. The title of the 2026 paper refers to a monoid

structure of records and a group structure of net balances, the distinction that this book develops in Part IV.

The two tools returned different sets. The CiNii search used one term, and only one of the three titles contains that exact term. The Scholar search used the author's name. A single query in a single database is a sample of the literature, and the log should record which query produced which result.

5.3. Reading a title in another language

If you cannot read the language of a paper, record the original title exactly as published and add your own translation in brackets, marked as yours. This book's bibliography follows that rule for every Japanese title. Do not romanize a title unless you have checked the reading, because a wrong reading makes the reference harder to find.

5.4. Exercises

1. Search CiNii Research for a term from your topic. Record how many results each record type (articles, books, dissertations) returned.
2. Search for the same term in Google Scholar. Which results appear in one tool and not the other?
3. For one paper you found, locate the repository record and write a complete reference for it.

6. Finding Older Literature

The oldest work in this book's bibliography was written in 1880. Finding and checking work that old took us different tools and more steps than finding recent papers.

6.1. Digitized books: the Internet Archive

The Internet Archive holds scanned copies of old books, including the one discussed in this section. It offers an advanced search that returns results as JSON, so a search can be logged exactly. We queried it for books with "philosophy of accounts" in the title and "Sprague" as creator. It returned eight records of *The Philosophy of Accounts* by Charles Ezra Sprague (1842–1912), dated 1908, 1920, 1922, and 1972. This book cites the 1908 scan (Sprague 1908).

When a book exists in several editions, record which edition you read. Page numbers and even content can change between editions.

6.2. A nineteenth-century article

Several modern papers cite a series of articles by Sprague titled "The Algebra of Accounts" from 1880. Our Internet Archive search did not return the 1880 issues of the periodical, but the articles were reprinted in a collection edited by Richard Brief, first published in 1982 and reissued in 2013 (Brief 2013). The Internet Archive's catalog record for the 1982 edition lists Sprague's "The algebra of accounts" among its contents. In the publisher's table of contents for the 2013 reissue, the chapter carries the note that it was written for *The Book-keeper*. This book cites the reprint (Sprague 2013).

The same publisher page also shows how catalog data can go wrong. It lists the chapter's author as "Charles E. Spraoue," credits two other chapters to the editor instead of their authors, and shows a description that belongs to a different book on depreciation. Take the facts you need from the reprinted text and from more than one record.

6.3. History papers as guides

Papers on the history of accounting can point to older primary sources. Two that our searches found have titles that match this book's themes:

- Peters and Emery (1978), on the role of negative numbers in the development of double-entry bookkeeping;
- Stone (2021), on the connection between Luca Pacioli, algebra, and double entry.

Part XIV returns to them.

6.4. Exercises

1. Search the Internet Archive for an old book related to your topic. Record the identifier and date of each edition it holds.
2. Find one reference in a modern paper that points to a source older than 1950. Try to locate the original. Record whether you succeeded.
3. Add an “open items” section to your research log, and list every reference you have not checked against an original.

7. Reading Papers

A search produces more papers than anyone can read in full. You need a way to decide quickly which papers matter, and a way to read the ones that do.

7.1. Read in passes

Keshav (2007) describes a practical three-pass method for reading research papers and explains how to use it for a literature survey. This book borrows the structure of reading a paper more than once, with each pass deciding whether the next is worth the time. Our passes are adapted for mathematical papers, and their details are ours.

1. **Survey.** Read the title, abstract, section headings, and conclusion. Decide whether the paper answers one of your questions from Chapter 1.
2. **Statements.** Read every definition and every theorem statement, and skip the proofs. Write each definition in your own notation.
3. **Proofs.** Read the proofs of the results you will use. Check each step, or formalize it.

The first pass exists to filter, so expect most papers to stop there.

7.2. Extracting the main claim

After the first pass, write the paper's main claim in one or two sentences, using only what the paper says. For Ellerman (2014) the abstract gives us the following.

- Double-entry bookkeeping implicitly uses a specific mathematical construction: the group of differences built from pairs of unsigned numbers, which the paper calls the Pacioli group.
- This construction was formulated abstractly in mathematics only in the nineteenth century, although bookkeeping had used it for centuries.
- A precise mathematical treatment yields a generalization of double entry to multi-dimensional vectors, which typically represent the different types of property in an enterprise or a household.

Each sentence above restates the abstract. None of them adds a claim the abstract does not make. When a summary goes beyond the source, mark the extra part as your own interpretation.

7.3. Reading a paper in a field you do not know

Papers on the algebra of bookkeeping come from accounting, mathematics, and computer science. A mathematician may not know what a "trial balance" is, and an accountant may not know what a "group of differences" is. Keep a two-column glossary in your log: the term as the paper uses it, and its meaning in the other field. Part II of this book introduces the bookkeeping terms, and the glossary in the appendix pairs them with their mathematical counterparts.

7.4. Exercises

1. Apply the first pass to five papers from your log. Record, for each, whether it answers one of your questions.
2. For one paper that passed, write its main claim in at most three sentences, using only what the paper says.
3. Choose one definition from that paper and rewrite it in your own notation.

8. Organizing What You Find

A search that is not recorded has to be repeated. This chapter covers three records worth keeping: the research log, the bibliography file, and a map of research lines.

8.1. The research log

The log records each search as it happens: date, tool, exact query, number of results, and the papers kept. It also has a place for references that are not yet checked (Chapter 6).

8.2. The bibliography file

This book stores references in a BibTeX file, `references.bib`. Quarto, the system used to build this book, generates citations and the bibliography through Pandoc. It needs a document with citations, a bibliographic data source such as a BibLaTeX or BibTeX file, and, optionally, a CSL file that sets the citation style. A document names its bibliography in its YAML header, for example `bibliography: references.bib`, and cites an entry by its key, for example `[@ellerman2014]`.

Each entry in this book's file was checked before it was added (Chapter 3). The file contains only bibliographic fields: authors, title, venue, volume, issue, pages, year, and DOI. A field that we could not check is left out, even when a search snippet showed a value for it.

8.3. A map of research lines

Grouping the papers by mathematical approach shows how the field divides. Here are the references in this book's bibliography, grouped that way. The later parts of this book follow the same lines.

Approach	Works	Years
Groups and the group of differences	Ellerman (1985), Ellerman (1986), Ellerman (2014)	1985–2014

Approach	Works	Years
Algebraic models and automata	Nehmer and Robinson (1997), Rambaud and Pérez (2005), Rambaud et al. (2010)	1997–2010
Matrices and linear algebra	Leech (1986), Arya et al. (2000), Arya et al. (2004)	1986–2004
Graphs	Liang (2023)	2023
Category theory	Katis et al. (2008)	2008 (written 1998)
Axioms and extensions of double entry	Ijiri (1965), Ijiri (1986)	1965–1986
Monoids, groups, and matrices in Japanese	Ozawa (2023a), Ozawa (2023b), Ozawa (2026)	2023–2026
History and early theory	Sprague (2013), Sprague (1908), Peters and Emery (1978), Stone (2021)	1880–2021

A paper can belong to more than one line. Ellerman (2014), for example, describes the group of differences and, according to its abstract, generalizes it to vectors. Put each paper where its main contribution lies, and note the other lines in the log.

8.4. Exercises

1. Create a BibTeX file for your topic. Add five entries, each checked against a publisher page, a registry, or an archive record.
2. Write a short Quarto document that cites all five entries, and render it. Fix any entry that fails to render.
3. Group the papers in your log into research lines, and draw the table for your topic.

9. Limits and Pitfalls

Several tools in Part 0 failed us at least once. This chapter records those failures, because a reader who repeats our search may meet the same ones.

9.1. Usage limits

Connected Papers. The free plan allows five graphs per month; the paid plans allow unlimited graphs. In our session, the first two graphs loaded. When we then requested graphs for Arya et al. (2000) and for a work titled *Momentum Accounting and Triple Entry Bookkeeping: Exploring the Dynamic Structure of Accounting Measurements*, the page asked us to log in and reported that the monthly graph limit had been reached. We did not see those two graphs.

Semantic Scholar API. Unauthenticated requests share one rate limit across all unauthenticated users (Chapter 4). Our unauthenticated search requests on 15 October 2026 were refused with the message "Too Many Requests," which suggests requesting an API key.

Crossref API. Some of our requests from the browser were refused with HTTP status 429, which signals too many requests. The same queries succeeded later from the command line.

Record each limit in the log, then continue with another tool. After the Semantic Scholar refusal we continued with Google Scholar.

9.2. Gaps in documentation

We looked for a description of Google Scholar's `author:` search operator in its search help page and did not find one there. Part 0 therefore teaches only the Scholar features we found documented: sorting by date, "Cited by," "Related articles," "All versions," and following a paper's references.

9.3. A registry match that was a different record

When we sent the citation of Ijiri's 1965 paper to Crossref, the best match had the right title and author but was a 2018 reprint in the journal *Accounting, Economics, and Law:*

A Convivium. A reader who copied that record would cite the wrong year and the wrong journal. Restricting the query to works published before 1967 returned the original in *The Accounting Review* (Ijiri 1965), but that record spells the author's name "Tjiri." Compare every field of a registry match with your source before you accept it (Chapter 3).

9.4. Copies from sites that are not the publisher

General web searches return PDFs from many kinds of sites. Our second web query returned a PDF of a book on accounting history from a document-sharing site. We could not establish that the site had the right to distribute the book, so this book does not use that copy. Use the publisher's version, an author's copy, a preprint server, or a library.

9.5. Fabricated references from chatbots

Walters and Wilder (2023) asked ChatGPT-3.5 and ChatGPT-4 to write short literature reviews on 42 topics and checked the 636 references in the 84 resulting papers. They report that 55% of the GPT-3.5 references and 18% of the GPT-4 references were fabricated, that is, they did not correspond to any real work. Among the references to real works, 43% from GPT-3.5 and 24% from GPT-4 contained substantive citation errors.

A language model can help to brainstorm search terms. It cannot replace a check against the source. Every reference in this book was checked against the source, a registry record, or an archive record.

9.6. Our own near-misses

The rules in this part come partly from mistakes we caught while writing it.

- We first wrote that a search restricted to the English term "double-entry" would miss the category theory paper Katis et al. (2008). The Scholar search log showed that a query containing "double-entry" did return it. We corrected the text in Chapter 2.
- We first romanized the Japanese titles in the bibliography without checking their readings. We replaced the romanizations with the original titles and marked our translations (Chapter 5).
- We first listed Nehmer and Robinson (1997) among the neighbors of Ellerman (2014) in the Connected Papers graph. It appeared in a different graph. We corrected Chapter 4.

We caught each error by comparing the text with the records of our searches.

9.7. Exercises

1. Record every usage limit you meet during your own search, with the date and the exact message.
2. Send three citations from your bibliography to Crossref. For each, compare every field of the best match with your source.
3. Ask a chatbot for five references on your topic. Check each one, and record which ones exist and which contain errors.

10. Writing Up the Survey

The last step turns the log into something another reader can use. This chapter shows the summary we wrote for this book.

10.1. A timeline

Sorting the checked references by year shows when each line of work appeared.

Year	Work	Line
1880	Sprague (2013)	Early theory
1908	Sprague (1908)	History
1965	Ijiri (1965)	Axioms
1978	Peters and Emery (1978)	History
1985	Ellerman (1985)	Groups
1986	Ellerman (1986), Leech (1986), Ijiri (1986)	Groups, matrices, extensions
1997	Nehmer and Robinson (1997)	Algebraic models
2000	Arya et al. (2000)	Linear algebra
2004	Arya et al. (2004)	Linear algebra
2005	Rambaud and Pérez (2005)	Automata
2008	Katis et al. (2008) (written 1998)	Category theory
2010	Rambaud et al. (2010)	Algebraic models
2014	Ellerman (2014)	Groups
2021	Stone (2021)	History
2023	Liang (2023), Ozawa (2023a), Ozawa (2023b)	Graphs, matrices
2026	Ozawa (2026)	Monoids and groups

The table lists only the works in this book's bibliography. It is not a complete history of the field.

10.2. Where this book stands

The works in the timeline apply different areas of mathematics to bookkeeping: groups, algebraic models and automata, matrices, graphs, and categories. Part 0 shows that each of these areas has been applied to bookkeeping before. This book takes them one at a time, on a single running example, and checks its main theorems in Lean. A survey can show that a piece of work exists; a search that finds nothing is weaker evidence, because the work may exist under words we did not search for.

10.3. Exercises

1. Build a timeline of the references in your log.
2. Write one paragraph that places your own project relative to the work you found. State what your search did not find, and how strong that evidence is.
3. For each claim in your paragraph, name the source in your log that supports it.

Part II.

Part I: The Language of Mathematics and Proof

Part I introduces the logic, sets, maps, and relations used in the rest of the book, each with a bookkeeping example.

11. Propositions and Logic

Level: ★

> **Note**
>
> This chapter is a placeholder. The sections below list what it will cover.

11.1. Planned sections

- Propositions and truth values
- Connectives: and, or, not, implies; truth tables
- Quantifiers: for all, there exists; negating quantified statements
- Bookkeeping example: stating the balance rule as a quantified proposition

11.2. Exercises

- To be written.

11.3. Formalization

- The Lean file for this chapter will be added under `lean/BokiAlgebra/`.

12. Writing Proofs

Level: ★

> **Note**
>
> This chapter is a placeholder. The sections below list what it will cover.

12.1. Planned sections

- Direct proof, contrapositive, contradiction, and proof by cases
- How to read and write a proof
- A first proof in Lean: definitions, `theorem`, and simple tactics

12.2. Exercises

- To be written.

12.3. Formalization

- The Lean file for this chapter will be added under `lean/BokiAlgebra/`.

13. Sets

Level: ★

> **Note**
>
> This chapter is a placeholder. The sections below list what it will cover.

13.1. Planned sections

- Sets, subsets, and power sets
- Union, intersection, difference, and complement
- Cartesian products
- Bookkeeping example: the set of accounts and the set of transactions

13.2. Exercises

- To be written.

13.3. Formalization

- The Lean file for this chapter will be added under `lean/BokiAlgebra/`.

14. Maps

Level: ★

> **Note**
>
> This chapter is a placeholder. The sections below list what it will cover.

14.1. Planned sections

- Maps, domain, codomain, image, preimage
- Injective, surjective, and bijective maps
- Composition and inverse maps
- Bookkeeping example: a journal entry as a map from accounts to amounts

14.2. Exercises

- To be written.

14.3. Formalization

- The Lean file for this chapter will be added under `lean/BokiAlgebra/`.

15. Relations and Equivalence Relations

Level: ★

> **Note**
>
> This chapter is a placeholder. The sections below list what it will cover.

15.1. Planned sections

- Relations and their properties
- Equivalence relations, equivalence classes, and quotient sets
- Bookkeeping example: two records that produce the same balance

15.2. Exercises

- To be written.

15.3. Formalization

- The Lean file for this chapter will be added under `lean/BokiAlgebra/`.

16. Orders

Level: ★★

> **Note**
>
> This chapter is a placeholder. The sections below list what it will cover.

16.1. Planned sections

- Partial orders and total orders
- Upper and lower bounds, suprema and infima
- Bookkeeping example: the hierarchy of a chart of accounts

16.2. Exercises

- To be written.

16.3. Formalization

- The Lean file for this chapter will be added under `lean/BokiAlgebra/`.

17. Induction and Recursion

Level: ★

> **Note**
>
> This chapter is a placeholder. The sections below list what it will cover.

17.1. Planned sections

- Mathematical induction and strong induction
- Recursive definitions
- Bookkeeping example: the trial balance stays balanced after any number of balanced entries

17.2. Exercises

- To be written.

17.3. Formalization

- The Lean file for this chapter will be added under `lean/BokiAlgebra/`.

18. Number Systems

Level: ★★

> **Note**
>
> This chapter is a placeholder. The sections below list what it will cover.

18.1. Planned sections

- Natural numbers, integers, rationals, and reals: an outline of their construction
- Integers built from pairs of natural numbers
- Bookkeeping example: the T-account as a pair of unsigned amounts

18.2. Exercises

- To be written.

18.3. Formalization

- The Lean file for this chapter will be added under `lean/BokiAlgebra/`.

Part III.

Part II: Bookkeeping Basics

Part II teaches the bookkeeping that the rest of the book formalizes. It uses no mathematics beyond arithmetic.

19. What Bookkeeping Is

Level: ★

> **Note**
>
> This chapter is a placeholder. The sections below list what it will cover.

19.1. Planned sections

- Recording, calculating, and reporting
- Single entry and double entry

19.2. Exercises

- To be written.

20. The Accounting Equation

Level: ★

> **Note**
>
> This chapter is a placeholder. The sections below list what it will cover.

20.1. Planned sections

- Assets, liabilities, and equity
- Revenue, expenses, and changes in equity

20.2. Exercises

- To be written.

21. Transactions and Journal Entries

Level: ★

> **Note**
>
> This chapter is a placeholder. The sections below list what it will cover.

21.1. Planned sections

- What counts as a transaction
- Debits and credits
- The journal

21.2. Exercises

- To be written.

22. Accounts and Posting

Level: ★

> **Note**
>
> This chapter is a placeholder. The sections below list what it will cover.

22.1. Planned sections

- T-accounts
- The general ledger
- Posting from the journal to the ledger

22.2. Exercises

- To be written.

23. The Trial Balance

Level: ★

> **Note**
>
> This chapter is a placeholder. The sections below list what it will cover.

23.1. Planned sections

- Totals and balances
- Errors a trial balance detects and errors it cannot detect

23.2. Exercises

- To be written.

24. Closing the Books

Level: ★

> **Note**
>
> This chapter is a placeholder. The sections below list what it will cover.

24.1. Planned sections

- Year-end adjustments: cost of goods sold, depreciation, allowances, accruals and deferrals
- The worksheet and closing entries

24.2. Exercises

- To be written.

25. Financial Statements

Level: ★

> **Note**
>
> This chapter is a placeholder. The sections below list what it will cover.

25.1. Planned sections

- The balance sheet
- The income statement
- An introduction to the cash flow statement

25.2. Exercises

- To be written.

26. The Running Example

Level: ★

> **Note**
>
> This chapter is a placeholder. The sections below list what it will cover.

26.1. Planned sections

- One company's transactions for one year
- Its journal, ledger, trial balance, and statements
- How later parts reuse this example

26.2. Exercises

- To be written.

Part IV.

Part III: Monoids and Groups

Part III introduces monoids and groups and uses them to describe journals and balances.

27. Binary Operations

Level: ★

> **Note**
>
> This chapter is a placeholder. The sections below list what it will cover.

27.1. Planned sections

- Binary operations, associativity, commutativity
- Identity and inverse elements
- Bookkeeping example: adding amounts and combining accounts

27.2. Exercises

- To be written.

27.3. Formalization

- The Lean file for this chapter will be added under `lean/BokiAlgebra/`.

28. Monoids

Level: ★

> **Note**
>
> This chapter is a placeholder. The sections below list what it will cover.

28.1. Planned sections

- Definition and examples
- Free monoids
- Bookkeeping example: the journal as a sequence of entries

28.2. Exercises

- To be written.

28.3. Formalization

- The Lean file for this chapter will be added under `lean/BokiAlgebra/`.

29. Groups

Level: ★

> **Note**
>
> This chapter is a placeholder. The sections below list what it will cover.

29.1. Planned sections

- Definition and examples
- Abelian groups
- The additive groups of integers, rationals, and reals

29.2. Exercises

- To be written.

29.3. Formalization

- The Lean file for this chapter will be added under `lean/BokiAlgebra/`.

30. Subgroups and Cosets

Level: ★★

> **Note**
>
> This chapter is a placeholder. The sections below list what it will cover.

30.1. Planned sections

- Subgroups
- Cosets and Lagrange's theorem

30.2. Exercises

- To be written.

30.3. Formalization

- The Lean file for this chapter will be added under `lean/BokiAlgebra/`.

31. Homomorphisms

Level: ★

> **Note**
>
> This chapter is a placeholder. The sections below list what it will cover.

31.1. Planned sections

- Homomorphisms, kernel, and image
- Isomorphisms
- Bookkeeping example: computing balances from the journal

31.2. Exercises

- To be written.

31.3. Formalization

- The Lean file for this chapter will be added under `lean/BokiAlgebra/`.

32. Quotient Groups

Level: ★★

> **Note**
>
> This chapter is a placeholder. The sections below list what it will cover.

32.1. Planned sections

- Normal subgroups and quotient groups
- The first isomorphism theorem
- Bookkeeping example: merging subsidiary accounts into a control account

32.2. Exercises

- To be written.

32.3. Formalization

- The Lean file for this chapter will be added under `lean/BokiAlgebra/`.

33. Direct Products and Direct Sums

Level: ★★

> **Note**
>
> This chapter is a placeholder. The sections below list what it will cover.

33.1. Planned sections

- Direct products and direct sums
- Universal properties
- Bookkeeping example: several currencies and several goods

33.2. Exercises

- To be written.

33.3. Formalization

- The Lean file for this chapter will be added under `lean/BokiAlgebra/`.

34. Free Abelian Groups

Level: ★★

> **Note**
>
> This chapter is a placeholder. The sections below list what it will cover.

34.1. Planned sections

- Bases and free abelian groups
- Bookkeeping example: accounts as a basis for the space of entries

34.2. Exercises

- To be written.

34.3. Formalization

- The Lean file for this chapter will be added under `lean/BokiAlgebra/`.

Part V.

Part IV: The Group of Differences

Part IV is the core of the book. It builds the group of differences from T-accounts and uses it to state and prove the rules of double entry.

35. T-Accounts as Pairs

Level: ★

> **Note**
>
> This chapter is a placeholder. The sections below list what it will cover.

35.1. Planned sections

- A T-account as a pair (debit total, credit total)
- Adding pairs
- When two pairs describe the same balance

35.2. Exercises

- To be written.

35.3. Formalization

- The Lean file for this chapter will be added under `lean/BokiAlgebra/`.

36. The Pacioli Group

Level: ★

> **Note**
>
> This chapter is a placeholder. The sections below list what it will cover.

36.1. Planned sections

- The group of differences on pairs of unsigned numbers
- Proof that it is a group
- Isomorphism with the integers and the reals

36.2. Exercises

- To be written.

36.3. Formalization

- The Lean file for this chapter will be added under `lean/BokiAlgebra/`.

36.4. Further reading

- Ellerman (1985); Ellerman (2014)

37. Journal Entries and the Trial Balance

Level: ★

> **Note**
>
> This chapter is a placeholder. The sections below list what it will cover.

37.1. Planned sections

- A journal entry as a map from accounts to the Pacioli group
- Balanced entries
- Theorem: a sum of balanced entries is balanced

37.2. Exercises

- To be written.

37.3. Formalization

- The Lean file for this chapter will be added under `lean/BokiAlgebra/`.

37.4. Further reading

- Ellerman (2014)

38. The Accounting Equation, Algebraically

Level: ★

> **Note**
>
> This chapter is a placeholder. The sections below list what it will cover.

38.1. Planned sections

- The balance sheet as a sum of balanced entries
- Closing revenue and expense accounts into equity

38.2. Exercises

- To be written.

38.3. Formalization

- The Lean file for this chapter will be added under `lean/BokiAlgebra/`.

39. The Grothendieck Group

Level: ★★

> **Note**
>
> This chapter is a placeholder. The sections below list what it will cover.

39.1. Planned sections

- Building a group from a commutative monoid
- The Pacioli group as a Grothendieck group
- The universal property

39.2. Exercises

- To be written.

39.3. Formalization

- The Lean file for this chapter will be added under `lean/BokiAlgebra/`.

40. The Monoid of Records and the Group of Balances

Level: ★★

> **Note**
>
> This chapter is a placeholder. The sections below list what it will cover.

40.1. Planned sections

- Accumulating records as a commutative monoid
- Net balances as an abelian group
- Totals trial balance and balances trial balance

40.2. Exercises

- To be written.

40.3. Formalization

- The Lean file for this chapter will be added under `lean/BokiAlgebra/`.

40.4. Further reading

- Ozawa (2026)

41. Multi-Dimensional Bookkeeping

Level: ★★

> **Note**
>
> This chapter is a placeholder. The sections below list what it will cover.

41.1. Planned sections

- Pairs of vectors
- Several kinds of property in one ledger

41.2. Exercises

- To be written.

41.3. Formalization

- The Lean file for this chapter will be added under `lean/BokiAlgebra/`.

41.4. Further reading

- Ellerman (1986); Ellerman (2014)

42. An Algebra of Errors

Level: ★★

> **Note**
>
> This chapter is a placeholder. The sections below list what it will cover.

42.1. Planned sections

- Errors a trial balance detects, described with kernels
- Reversing entries and correcting entries

42.2. Exercises

- To be written.

42.3. Formalization

- The Lean file for this chapter will be added under `lean/BokiAlgebra/`.

Part VI.

Part V: Linear Algebra

Part V writes balances as vectors and entries as matrices, and uses linear algebra to analyze them.

43. Vector Spaces

Level: ★

> **Note**
>
> This chapter is a placeholder. The sections below list what it will cover.

43.1. Planned sections

- Vector spaces and subspaces
- Linear independence, bases, and dimension
- Bookkeeping example: balance vectors indexed by accounts

43.2. Exercises

- To be written.

43.3. Formalization

- The Lean file for this chapter will be added under `lean/BokiAlgebra/`.

44. Matrices and Linear Maps

Level: ★

> **Note**
>
> This chapter is a placeholder. The sections below list what it will cover.

44.1. Planned sections

- Matrix multiplication
- The matrix of a linear map
- Bookkeeping example: a journal entry as a matrix

44.2. Exercises

- To be written.

44.3. Formalization

- The Lean file for this chapter will be added under `lean/BokiAlgebra/`.

45. Matrix Bookkeeping

Level: ★

> **Note**
>
> This chapter is a placeholder. The sections below list what it will cover.

45.1. Planned sections

- Transaction matrices with debit accounts as rows and credit accounts as columns
- The trial balance from row sums and column sums

45.2. Exercises

- To be written.

45.3. Formalization

- The Lean file for this chapter will be added under `lean/BokiAlgebra/`.

45.4. Further reading

- Leech (1986); Ozawa (2023a)

46. Kernel, Image, and Rank

Level: ★

> **Note**
>
> This chapter is a placeholder. The sections below list what it will cover.

46.1. Planned sections

- Kernel, image, and the rank-nullity theorem
- Balanced entries as the subspace where the components sum to zero

46.2. Exercises

- To be written.

46.3. Formalization

- The Lean file for this chapter will be added under `lean/BokiAlgebra/`.

47. Systems of Linear Equations

Level: ★

> **Note**
>
> This chapter is a placeholder. The sections below list what it will cover.

47.1. Planned sections

- Gaussian elimination
- Existence and uniqueness of solutions
- Recovering transactions from financial statements

47.2. Exercises

- To be written.

47.3. Formalization

- The Lean file for this chapter will be added under `lean/BokiAlgebra/`.

48. Least Squares and Generalized Inverses

Level: ★★

> **Note**
>
> This chapter is a placeholder. The sections below list what it will cover.

48.1. Planned sections

- Normal equations and projections
- The Moore-Penrose inverse
- Estimating transactions from statements

48.2. Exercises

- To be written.

48.3. Formalization

- The Lean file for this chapter will be added under `lean/BokiAlgebra/`.

48.4. Further reading

- Arya et al. (2000)

49. Inner Products and Duality

Level: ★★

> **Note**
>
> This chapter is a placeholder. The sections below list what it will cover.

49.1. Planned sections

- Inner products and orthogonal complements
- Dual spaces
- Bookkeeping example: prices as linear functionals that turn quantities into amounts

49.2. Exercises

- To be written.

49.3. Formalization

- The Lean file for this chapter will be added under `lean/BokiAlgebra/`.

50. Aggregation and Projection

Level: ★★

> **Note**
>
> This chapter is a placeholder. The sections below list what it will cover.

50.1. Planned sections

- Linear maps that merge accounts
- Reconciling information at different levels of aggregation

50.2. Exercises

- To be written.

50.3. Formalization

- The Lean file for this chapter will be added under `lean/BokiAlgebra/`.

50.4. Further reading

- Arya et al. (2004)

51. Eigenvalues and Diagonalization

Level: ★★

> **Note**
>
> This chapter is a placeholder. The sections below list what it will cover.

51.1. Planned sections

- Eigenvalues and eigenvectors
- Diagonalization
- Bookkeeping example: repeated transactions

51.2. Exercises

- To be written.

51.3. Formalization

- The Lean file for this chapter will be added under `lean/BokiAlgebra/`.

52. Input-Output Analysis

Level: ★★★

> **Note**
>
> This chapter is a placeholder. The sections below list what it will cover.

52.1. Planned sections

- The Leontief matrix and its inverse
- Allocating service department costs as a linear system

52.2. Exercises

- To be written.

52.3. Formalization

- The Lean file for this chapter will be added under `lean/BokiAlgebra/`.

Part VII.

Part VI: Rings, Modules, and Order

Part VI adds rings, modules, and ordered structures.

53. Rings and Fields

Level: ★★

> **Note**
>
> This chapter is a placeholder. The sections below list what it will cover.

53.1. Planned sections

- Rings, fields, and an introduction to ideals

53.2. Exercises

- To be written.

53.3. Formalization

- The Lean file for this chapter will be added under `lean/BokiAlgebra/`.

54. Modules

Level: ★★

> **Note**
>
> This chapter is a placeholder. The sections below list what it will cover.

54.1. Planned sections

- Modules over a ring
- Abelian groups as modules over the integers
- Quantities over the integers and amounts over the reals

54.2. Exercises

- To be written.

54.3. Formalization

- The Lean file for this chapter will be added under `lean/BokiAlgebra/`.

54.4. Further reading

- Rambaud et al. (2010)

55. Tensor Products

Level: ★★★

> **Note**
>
> This chapter is a placeholder. The sections below list what it will cover.

55.1. Planned sections

- Tensor products of modules
- Bookkeeping example: combining quantities and unit prices

55.2. Exercises

- To be written.

55.3. Formalization

- The Lean file for this chapter will be added under `lean/BokiAlgebra/`.

56. Ordered Groups

Level: ★★

> **Note**
>
> This chapter is a placeholder. The sections below list what it will cover.

56.1. Planned sections

- Ordered groups and positive cones
- Unsigned debit and credit amounts as an ordered structure

56.2. Exercises

- To be written.

56.3. Formalization

- The Lean file for this chapter will be added under `lean/BokiAlgebra/`.

57. Lattices and the Chart of Accounts

Level: ★★

> **Note**
>
> This chapter is a placeholder. The sections below list what it will cover.

57.1. Planned sections

- Lattices and Boolean algebras
- Consistent aggregation in a chart of accounts

57.2. Exercises

- To be written.

57.3. Formalization

- The Lean file for this chapter will be added under `lean/BokiAlgebra/`.

Part VIII.

Part VII: Graphs and Networks

Part VII draws accounts as vertices and transactions as edges.

58. Graphs

Level: ★

> **Note**
>
> This chapter is a placeholder. The sections below list what it will cover.

58.1. Planned sections

- Directed graphs, multigraphs, paths, and connectivity
- The bookkeeping graph: accounts as vertices, transactions as edges

58.2. Exercises

- To be written.

58.3. Formalization

- The Lean file for this chapter will be added under `lean/BokiAlgebra/`.

59. The Incidence Matrix

Level: ★

> **Note**
>
> This chapter is a placeholder. The sections below list what it will cover.

59.1. Planned sections

- The incidence matrix of a directed graph
- Its column sums
- The trial balance and Kirchhoff's current law

59.2. Exercises

- To be written.

59.3. Formalization

- The Lean file for this chapter will be added under `lean/BokiAlgebra/`.

60. Flows and Conservation

Level: ★★

> **Note**
>
> This chapter is a placeholder. The sections below list what it will cover.

60.1. Planned sections

- Network flows and flow conservation
- Cash flows on the bookkeeping graph

60.2. Exercises

- To be written.

60.3. Formalization

- The Lean file for this chapter will be added under `lean/BokiAlgebra/`.

61. Trees and Spanning Trees

Level: ★★

> **Note**
>
> This chapter is a placeholder. The sections below list what it will cover.

61.1. Planned sections

- Trees and spanning trees
- The matrix-tree theorem
- The account hierarchy as a tree

61.2. Exercises

- To be written.

61.3. Formalization

- The Lean file for this chapter will be added under `lean/BokiAlgebra/`.

62. Cycle Space and Cut Space

Level: ★★

> **Note**
>
> This chapter is a placeholder. The sections below list what it will cover.

62.1. Planned sections

- Kernel and image of the incidence matrix
- Transactions that return money to where it started

62.2. Exercises

- To be written.

62.3. Formalization

- The Lean file for this chapter will be added under `lean/BokiAlgebra/`.

63. An Introduction to Graph Homology

Level: ★★★

> **Note**
>
> This chapter is a placeholder. The sections below list what it will cover.

63.1. Planned sections

- Chain complexes, boundary maps, and homology groups
- Bookkeeping in the language of homology

63.2. Exercises

- To be written.

63.3. Formalization

- The Lean file for this chapter will be added under `lean/BokiAlgebra/`.

64. Bookkeeping Graphs

Level: ★★★

> **Note**
>
> This chapter is a placeholder. The sections below list what it will cover.

64.1. Planned sections

- Computational theory of bookkeeping graphs

64.2. Exercises

- To be written.

64.3. Formalization

- The Lean file for this chapter will be added under `lean/BokiAlgebra/`.

64.4. Further reading

- Liang (2023)

Part IX.

Part VIII: Category Theory

Part VIII introduces categories, functors, and universal properties, and returns to earlier constructions in that language.

65. Categories

Level: ★★

> **Note**
>
> This chapter is a placeholder. The sections below list what it will cover.

65.1. Planned sections

- Objects, morphisms, composition, identities
- Examples: sets, groups, vector spaces
- A category with accounts as objects and transactions as morphisms

65.2. Exercises

- To be written.

65.3. Formalization

- The Lean file for this chapter will be added under `lean/BokiAlgebra/`.

66. Functors and Natural Transformations

Level: ★★

> **Note**
>
> This chapter is a placeholder. The sections below list what it will cover.

66.1. Planned sections

- Functors and forgetful functors
- Natural transformations
- From books to statements as a functor

66.2. Exercises

- To be written.

66.3. Formalization

- The Lean file for this chapter will be added under `lean/BokiAlgebra/`.

67. Universal Properties

Level: ★★

> **Note**
>
> This chapter is a placeholder. The sections below list what it will cover.

67.1. Planned sections

- Products, coproducts, initial and terminal objects
- The Grothendieck group by its universal property

67.2. Exercises

- To be written.

67.3. Formalization

- The Lean file for this chapter will be added under `lean/BokiAlgebra/`.

68. Adjunctions

Level: ★★★

> **Note**
>
> This chapter is a placeholder. The sections below list what it will cover.

68.1. Planned sections

- Free and forgetful adjunctions
- Building the Grothendieck group as an adjunction

68.2. Exercises

- To be written.

68.3. Formalization

- The Lean file for this chapter will be added under `lean/BokiAlgebra/`.

69. Monoidal Categories and String Diagrams

Level: ★★★

> **Note**
>
> This chapter is a placeholder. The sections below list what it will cover.

69.1. Planned sections

- Monoidal categories
- Drawing string diagrams
- Transactions as string diagrams

69.2. Exercises

- To be written.

69.3. Formalization

- The Lean file for this chapter will be added under `lean/BokiAlgebra/`.

70. Spans and Cospans

Level: ★★★

> **Note**
>
> This chapter is a placeholder. The sections below list what it will cover.

70.1. Planned sections

- Spans, cospans, and their composition
- Partita doppia in terms of spans of reflexive graphs

70.2. Exercises

- To be written.

70.3. Formalization

- The Lean file for this chapter will be added under `lean/BokiAlgebra/`.

70.4. Further reading

- Katis et al. (2008)

Part X.

Part IX: Time and Change

Part IX adds time: sequences, present value, calculus, and probability.

71. Sequences and Difference Equations

Level: ★

> **Note**
>
> This chapter is a placeholder. The sections below list what it will cover.

71.1. Planned sections

- Recurrences for opening and closing balances

71.2. Exercises

- To be written.

71.3. Formalization

- The Lean file for this chapter will be added under `lean/BokiAlgebra/`.

72. Series and Present Value

Level: ★

> **Note**
>
> This chapter is a placeholder. The sections below list what it will cover.

72.1. Planned sections

- Geometric series, discounting, and annuities
- Depreciation methods as sequences

72.2. Exercises

- To be written.

72.3. Formalization

- The Lean file for this chapter will be added under `lean/BokiAlgebra/`.

73. Derivatives and Integrals

Level: ★★

> **Note**
>
> This chapter is a placeholder. The sections below list what it will cover.

73.1. Planned sections

- Balances as functions of time: stocks and flows
- The fundamental theorem of calculus and the link between balances and transactions

73.2. Exercises

- To be written.

73.3. Formalization

- The Lean file for this chapter will be added under `lean/BokiAlgebra/`.

74. Differential Equations

Level: ★★

> **Note**
>
> This chapter is a placeholder. The sections below list what it will cover.

74.1. Planned sections

- Continuous compounding and growth models

74.2. Exercises

- To be written.

74.3. Formalization

- The Lean file for this chapter will be added under `lean/BokiAlgebra/`.

75. Triple-Entry Bookkeeping

Level: ★★

> **Note**
>
> This chapter is a placeholder. The sections below list what it will cover.

75.1. Planned sections

- Wealth, momentum, and force accounting
- How the layers relate over time

75.2. Exercises

- To be written.

75.3. Formalization

- The Lean file for this chapter will be added under `lean/BokiAlgebra/`.

75.4. Further reading

- Ijiri (1986)

76. Probability and Estimates

Level: ★★

> **Note**
>
> This chapter is a placeholder. The sections below list what it will cover.

76.1. Planned sections

- Random variables and expectation
- Estimating allowances

76.2. Exercises

- To be written.

76.3. Formalization

- The Lean file for this chapter will be added under `lean/BokiAlgebra/`.

Part XI.

Part X: Computation and Information

Part X treats books of account as programs and data.

77. Automata

Level: ★★

> **Note**
>
> This chapter is a placeholder. The sections below list what it will cover.

77.1. Planned sections

- Finite automata, states, and transitions
- An accounting system as an algebraic automaton

77.2. Exercises

- To be written.

77.3. Formalization

- The Lean file for this chapter will be added under `lean/BokiAlgebra/`.

77.4. Further reading

- Rambaud and Pérez (2005)

78. The Ledger as a Data Structure

Level: ★

> **Note**
>
> This chapter is a placeholder. The sections below list what it will cover.

78.1. Planned sections

- Data structures for journal entries
- Append-only records

78.2. Exercises

- To be written.

78.3. Formalization

- The Lean file for this chapter will be added under `lean/BokiAlgebra/`.

79. Relational Databases

Level: ★★

> **Note**
>
> This chapter is a placeholder. The sections below list what it will cover.

79.1. Planned sections

- Relational algebra and normalization
- Designing accounting data

79.2. Exercises

- To be written.

79.3. Formalization

- The Lean file for this chapter will be added under `lean/BokiAlgebra/`.

80. Plain-Text Accounting

Level: ★

> **Note**
>
> This chapter is a placeholder. The sections below list what it will cover.

80.1. Planned sections

- Ledgers kept as text files
- Checking the theorems of this book on a real ledger file

80.2. Exercises

- To be written.

80.3. Formalization

- The Lean file for this chapter will be added under `lean/BokiAlgebra/`.

81. Error Detection

Level: ★★

> **Note**
>
> This chapter is a placeholder. The sections below list what it will cover.

81.1. Planned sections

- Parity checks and error-correcting codes
- Double entry as redundancy for detecting errors

81.2. Exercises

- To be written.

81.3. Formalization

- The Lean file for this chapter will be added under `lean/BokiAlgebra/`.

82. Types That Rule Out Unbalanced Entries

Level: ★★★

> **Note**
>
> This chapter is a placeholder. The sections below list what it will cover.

82.1. Planned sections

- Type systems that make an unbalanced entry impossible to write

82.2. Exercises

- To be written.

82.3. Formalization

- The Lean file for this chapter will be added under `lean/BokiAlgebra/`.

Part XII.

Part XI: Formalizing Bookkeeping in Lean

Part XI collects the formalization sections of earlier chapters into one Lean library.

83. Lean 4 Basics

Level: ★

> **Note**
>
> This chapter is a placeholder. The sections below list what it will cover.

83.1. Planned sections

- Types, functions, theorems, and proofs
- Basic tactics: `intro`, `apply`, `rw`, `simp`, `ring`, `omega`

83.2. Exercises

- To be written.

83.3. Formalization

- The Lean file for this chapter will be added under `lean/BokiAlgebra/`.

84. Using Mathlib

Level: ★

> **Note**
>
> This chapter is a placeholder. The sections below list what it will cover.

84.1. Planned sections

- The algebraic hierarchy: `Monoid`, `AddCommGroup`, `Module`
- Finding lemmas with `exact?` and the documentation

84.2. Exercises

- To be written.

84.3. Formalization

- The Lean file for this chapter will be added under `lean/BokiAlgebra/`.

85. Defining Accounts and Journal Entries

Level: ★

> **Note**
>
> This chapter is a placeholder. The sections below list what it will cover.

85.1. Planned sections

- A type of accounts
- A structure for journal entries

85.2. Exercises

- To be written.

85.3. Formalization

- The Lean file for this chapter will be added under `lean/BokiAlgebra/`.

86. Building the Pacioli Group

Level: ★★

> **Note**
>
> This chapter is a placeholder. The sections below list what it will cover.

86.1. Planned sections

- The type of pairs, an equivalence relation, and the quotient type
- An `AddCommGroup` instance

86.2. Exercises

- To be written.

86.3. Formalization

- The Lean file for this chapter will be added under `lean/BokiAlgebra/`.

87. Proving That the Trial Balance Balances

Level: ★★

> **Note**
>
> This chapter is a placeholder. The sections below list what it will cover.

87.1. Planned sections

- The sum of balanced entries is balanced
- A proof by induction

87.2. Exercises

- To be written.

87.3. Formalization

- The Lean file for this chapter will be added under `lean/BokiAlgebra/`.

88. Comparing with the Grothendieck Group in Mathlib

Level: ★★★

> **Note**
>
> This chapter is a placeholder. The sections below list what it will cover.

88.1. Planned sections

- Mathlib's `GrothendieckGroup`
- An isomorphism with the Pacioli group

88.2. Exercises

- To be written.

88.3. Formalization

- The Lean file for this chapter will be added under `lean/BokiAlgebra/`.

89. Matrix Bookkeeping in Lean

Level: ★★★

> **Note**
>
> This chapter is a placeholder. The sections below list what it will cover.

89.1. Planned sections

- Transaction matrices with `Matrix`
- Row sums and column sums

89.2. Exercises

- To be written.

89.3. Formalization

- The Lean file for this chapter will be added under `lean/BokiAlgebra/`.

90. Graphs and Incidence Matrices in Lean

Level: ★★★

> **Note**
>
> This chapter is a placeholder. The sections below list what it will cover.

90.1. Planned sections

- Bookkeeping graphs and their incidence matrices

90.2. Exercises

- To be written.

90.3. Formalization

- The Lean file for this chapter will be added under `lean/BokiAlgebra/`.

91. Checking the Running Example in Lean

Level: ★★

> **Note**
>
> This chapter is a placeholder. The sections below list what it will cover.

91.1. Planned sections

- Entering one year of transactions
- Checking that the statements satisfy the accounting equation

91.2. Exercises

- To be written.

91.3. Formalization

- The Lean file for this chapter will be added under `lean/BokiAlgebra/`.

92. Running a Formalization Project

Level: ★★

> **Note**
>
> This chapter is a placeholder. The sections below list what it will cover.

92.1. Planned sections

- Lake, CI, and generated documentation

92.2. Exercises

- To be written.

92.3. Formalization

- The Lean file for this chapter will be added under `lean/BokiAlgebra/`.

Part XIII.

Part XII: Applications in Accounting

Part XII applies the mathematics of earlier parts to further accounting topics.

93. Consolidation

Level: ★★

> **Note**
>
> This chapter is a placeholder. The sections below list what it will cover.

93.1. Planned sections

- Eliminating intercompany transactions as a quotient
- Ownership calculations with linear algebra

93.2. Exercises

- To be written.

93.3. Formalization

- The Lean file for this chapter will be added under `lean/BokiAlgebra/`.

94. Foreign Currency Translation

Level: ★★

> **Note**
>
> This chapter is a placeholder. The sections below list what it will cover.

94.1. Planned sections

- Exchange rates as linear maps
- Multi-currency accounts and translation differences

94.2. Exercises

- To be written.

94.3. Formalization

- The Lean file for this chapter will be added under `lean/BokiAlgebra/`.

95. Cost Accounting

Level: ★★

> **Note**
>
> This chapter is a placeholder. The sections below list what it will cover.

95.1. Planned sections

- Departmental costing and service department allocation as linear systems
- Standard costing and variance analysis

95.2. Exercises

- To be written.

95.3. Formalization

- The Lean file for this chapter will be added under `lean/BokiAlgebra/`.

96. Management Accounting and Optimization

Level: ★★

> **Note**
>
> This chapter is a placeholder. The sections below list what it will cover.

96.1. Planned sections

- Break-even analysis
- Linear programming

96.2. Exercises

- To be written.

96.3. Formalization

- The Lean file for this chapter will be added under `lean/BokiAlgebra/`.

97. The Cash Flow Statement, Algebraically

Level: ★★

> **Note**
>
> This chapter is a placeholder. The sections below list what it will cover.

97.1. Planned sections

- Deriving the indirect cash flow statement from balance sheet differences

97.2. Exercises

- To be written.

97.3. Formalization

- The Lean file for this chapter will be added under `lean/BokiAlgebra/`.

97.4. Further reading

- Ozawa (2023b)

98. Auditing and Anomaly Detection

Level: ★★

> **Note**
>
> This chapter is a placeholder. The sections below list what it will cover.

98.1. Planned sections

- Statistical sampling
- Detecting unusual entries

98.2. Exercises

- To be written.

98.3. Formalization

- The Lean file for this chapter will be added under `lean/BokiAlgebra/`.

Part XIV.

Part XIII: Applications in Economics

Part XIII applies the same structures to accounts kept for whole economies.

99. National Accounts

Level: ★★

> **Note**
>
> This chapter is a placeholder. The sections below list what it will cover.

99.1. Planned sections

- The system of national accounts and entries between sectors

99.2. Exercises

- To be written.

99.3. Formalization

- The Lean file for this chapter will be added under `lean/BokiAlgebra/`.

100. Input-Output Tables

Level: ★★

> **Note**
>
> This chapter is a placeholder. The sections below list what it will cover.

100.1. Planned sections

- Input-output tables as matrices

100.2. Exercises

- To be written.

100.3. Formalization

- The Lean file for this chapter will be added under `lean/BokiAlgebra/`.

101. Social Accounting Matrices

Level: ★★★

> **Note**
>
> This chapter is a placeholder. The sections below list what it will cover.

101.1. Planned sections

- Social accounting matrices and equal row and column sums

101.2. Exercises

- To be written.

101.3. Formalization

- The Lean file for this chapter will be added under `lean/BokiAlgebra/`.

Part XV.

Part XIV: History and Research Directions

Part XIV places the mathematics of the book in its history and lists questions for further study.

102. The Origins of Double Entry

Level: ★

> **Note**
>
> This chapter is a placeholder. The sections below list what it will cover.

102.1. Planned sections

- Luca Pacioli and the connection between algebra and double entry

102.2. Exercises

- To be written.

102.3. Further reading

- Stone (2021)

103. Negative Numbers and Bookkeeping

Level: ★★

> **Note**
>
> This chapter is a placeholder. The sections below list what it will cover.

103.1. Planned sections

- The role of negative numbers in the development of double entry

103.2. Exercises

- To be written.

103.3. Further reading

- Peters and Emery (1978)

104. The Algebra of Accounts, from Sprague Onward

Level: ★★

> **Note**
>
> This chapter is a placeholder. The sections below list what it will cover.

104.1. Planned sections

- Sprague's algebra of accounts
- Axioms of accounting measurement
- The group of differences

104.2. Exercises

- To be written.

104.3. Further reading

- Sprague (2013); Sprague (1908); Ijiri (1965); Ellerman (1985)

105. Research in Japanese

Level: ★★

> **Note**
>
> This chapter is a placeholder. The sections below list what it will cover.

105.1. Planned sections

- Bookkeeping algebra models in Japanese research

105.2. Exercises

- To be written.

105.3. Further reading

- Ozawa (2023a); Ozawa (2023b); Ozawa (2026)

106. Open Problems and Thesis Topics

Level: ★★★

> **Note**
>
> This chapter is a placeholder. The sections below list what it will cover.

106.1. Planned sections

- Questions for further study
- Starting a survey with the methods of Part 0

106.2. Exercises

- To be written.

A. Notation

> **Note**
>
> This chapter is a placeholder. The sections below list what it will cover.

A.1. Planned sections

- Symbols used in the book

B. Glossary

> **Note**
>
> This chapter is a placeholder. The sections below list what it will cover.

B.1. Planned sections

- Bookkeeping terms and their mathematical counterparts, in English and Japanese

C. The Running Example in Full

> **Note**
>
> This chapter is a placeholder. The sections below list what it will cover.

C.1. Planned sections

- All transactions, the journal, the ledger, and the statements

D. Lean Tactic Reference

> **Note**
>
> This chapter is a placeholder. The sections below list what it will cover.

D.1. Planned sections

- The tactics used in the book

E. Solutions to Exercises

> **Note**
>
> This chapter is a placeholder. The sections below list what it will cover.

E.1. Planned sections

- Solutions, chapter by chapter

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